

Economic World Models: A Survey of Agent-Based and Learned Simulation in Financial Markets and Macroeconomics

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Abstract

Financial strategies and macroeconomic policies are difficult to test because historical backtests expose only realized trajectories, whereas field interventions are costly, weakly controlled, and capable of changing the system under study. Economic world models (EWMs) offer an interactive alternative: executable economies in which institutional state, heterogeneous actors, and learned decision or transition components evolve under counterfactual strategies, shocks, and policies. The relevant evidence, however, is divided among rule-based agent-based models (ABMs), data-driven generative markets, and large language model (LLM) agent societies.

This survey connects these three literatures across market microstructure, financial institutions, macroeconomics, and market-society systems. It makes three contributions. First, it defines the boundary between EWMs, conventional ABMs, passive forecasting, and historical replay. Second, it organizes representative systems through a two-dimensional taxonomy: economic level and the location of learning in agents, environments, or both. Third, it synthesizes evaluation practice around institutional and accounting integrity, behavioral realism, distributional fit, intervention validity, and reproducibility, and separates datasets from simulators and evaluation suites. We then distinguish experimentally supported uses in market and policy research from prospective banking and regulatory applications. The review finds complementary strengths but no all-purpose system, with behavioral identification, cross-regime transfer, accounting integration, LLM reliability, and computational governance remaining central challenges.

Keywords

economic world models, agent-based simulation, LLM agents, generative markets, macroeconomic ABMs, systemic risk simulation

1 Introduction

1.1 Research background

Financial and macroeconomic interventions are reflexive: they alter the systems in which they operate. A trading strategy changes order flow and market impact, a liquidity action changes beliefs and balance sheets, and a policy rule changes both behavior and the future data-generating process [63]. Historical replay observes only one realized path and usually holds market response fixed. Field experiments offer stronger evidence but are often expensive, ethically constrained, or infeasible at systemic scale. These limitations have long motivated computational economies built from heterogeneous actors and explicit institutions [6, 24, 26, 70].

Machine learning widens this experimental space. RL policies adapt inside simulated markets, generative models learn market response from event data, and LLM agents process news, communicate, and make household, firm, or policy decisions. These

developments motivate the emerging idea of an *economic world model*: an interactive computational economy whose endogenous responses can be examined under alternative actions and interventions [19, 33]. The value of an EWM is therefore not a more elaborate backtest, but a controlled environment in which actions have consequences for other actors and for subsequent economic state.

1.2 Fragmented research landscape

The literature has developed along three largely separate lines. *Rule-based ABMs* emphasize heterogeneous interaction, exchange institutions, balance sheets, and emergent outcomes. *Generative markets* learn order flow or state transitions from observed sequences, often without representing identifiable actors. *LLM multi-agent systems* emphasize cognition, memory, communication, and open-ended choice, but frequently use simplified economic institutions. This division produces different meanings of realism: mechanism fidelity in ABMs, distributional similarity in generators, and plausible behavior in LLM societies.

The survey literature reflects the same separation. Economic-ABM reviews emphasize emergence, calibration, forecasting, and central-bank use [6, 9, 57]. ICAIF and adjacent financial surveys cover capital-market algorithms, synthetic data, quantitative trading, financial LLMs, or investment agents [5, 20, 23, 41, 46, 64]. Surveys of financial multi-agent systems and generative social simulation move closer to interactive worlds, but do not jointly organize transaction mechanics, macroeconomic accounting, learned environments, and language-mediated behavior [42, 48]. Table 1 makes this gap explicit without treating topical coverage as a quality score.

1.3 Contributions of this survey

This survey makes three contributions:

- **Cross-level integration.** We connect market simulators, institution and network models, macroeconomic ABMs, generative markets, and LLM-agent societies under a bounded EWM concept.
- **Two-dimensional taxonomy.** We classify systems by economic level and by whether learning occurs in decision makers, the environment, or both, while retaining rule-based ABMs as institutional antecedents.
- **Evaluation synthesis.** We align validation dimensions, domain metrics, and available data or benchmark assets, then separate demonstrated experimental uses from prospective deployment claims.

Larooij and Törnberg establish purpose-conditioned operational validity as a central challenge for generative social simulation [42]. Our contribution is complementary rather than duplicative: it specializes that concern to financial transaction rules, institution and network state, macroeconomic accounts, learned market response,

Table 1: Organizing emphases of adjacent reviews. • denotes a substantive focus, ◦ adjacent coverage, and — a topic outside the stated scope; symbols do not rate review quality or rigor.

Review	Market mechanics	Macro accounts	Learned environment	LLM behavior	Policy response	Evaluation
Axtell–Farmer; Pangallo–del Río–Chanona [6, 57]	◦	•	◦	—	◦	•
Capital-market and trading surveys [23, 41]	•	—	◦	◦	—	◦
Financial LLM and investment-agent surveys [20, 46, 64]	◦	—	—	•	—	◦
Synthetic-finance review [5]	◦	—	•	—	—	◦
Financial LLM–MAS survey [48]	•	◦	◦	•	◦	◦
Generative-social-simulation review [42]	—	—	◦	•	◦	•

and intervention claims. The result is a comparative survey, not a new validation algorithm or a deployment-maturity scale.

1.4 Literature scope and selection

We use a structured narrative-review approach rather than claim an exhaustive systematic review or meta-analysis. A reproducible OpenAlex title-and-abstract search, closed on July 26, 2026, used ten query families spanning EWMs, financial and macroeconomic ABMs, learned markets, financial digital twins, LLM-agent simulation, and systemic risk. It returned 308 query hits and 253 records after OpenAlex-ID and normalized title–year deduplication; these are discovery, not inclusion, counts. We supplemented this set with targeted ACM Digital Library, DBLP, and ICAIF 2020–2025 proceedings checks and backward and forward citation chaining. Archival versions were preferred; recent preprints or institutional projects were retained when they introduced an inspectable system or evaluation setting not represented by an archival study. Focal systems required an evolving economic state, endogenous interaction or response, reported economic outputs, and inspectable methods or results; validation and governance sources were tracked as contextual evidence.

1.5 Organization

Section 2 defines EWMs, traces their three technical lineages, and introduces the taxonomy. Section 3 reviews systems from market microstructure to market–society simulation. Section 4 synthesizes validation, metrics, and benchmark resources. Section 5 separates demonstrated experimental uses from prospective applications, Section 6 identifies common limitations and research tracks, and Sections 7 and 8 state the survey’s limitations and conclusions.

2 EWM Definition and Technical Lineages

2.1 A strict working definition

We use *EWM* for an interactive simulator with four jointly required properties: (i) an *executable economic state*, such as an order book, portfolio and balance-sheet network, or sectoral account system; (ii) *endogenous interaction*, so an actor’s decision changes prices, allocations, constraints, or information faced by others; (iii) an *intervention interface* through which a strategy, shock, rule, or policy changes the rollout; and (iv) an *embedded learned component* that affects actor decisions, institutional policy, or state transitions during rollout or adaptation. This is a systems-level review definition, not a replacement for ABM and not the internal latent dynamics model commonly called a world model in model-based reinforcement learning.

A rule-based ABM may satisfy the first three properties and provide the indispensable institutional substrate for an EWM, but it falls outside the core boundary until a learned rollout component is added. Conversely, a static backtest, passive predictor, fixed replay, single agent acting against non-responsive history, or persona collection without a consequential shared state is not an EWM. Learning used only for offline calibration, initialization, or state estimation supports an EWM but does not by itself satisfy the fourth property. Stock–flow accounting is required for economy-wide balance-sheet claims, although not for a venue-level execution experiment. The three lineages below contribute institutional mechanisms, learned response, and language-mediated behavior to this definition.

2.2 Three technical lineages

Agent-based computational economics. ACE explains aggregate outcomes through heterogeneous, boundedly rational actors rather than a representative-agent solution [24, 70]. Financial ABMs make order submission, price formation, leverage, and contagion endogenous; macro ABMs connect households, firms, banks, government, and central banks through transactions and accounts [25, 26]. Their main contribution is executable institutional structure.

Data-driven generative simulation. Calibration aligns parameters or summaries with data, data assimilation updates latent state, and learned transition models replace selected hand-coded mechanisms [32, 60, 75]. Bayesian optimization, gradient-assisted inference, and RL search reduce the cost of fitting high-dimensional ABMs [7, 22, 31]. Generative markets go further: they learn how order flow or market state responds to history and incoming actions, turning observations into a responsive environment rather than a fixed replay [17, 44].

LLM language agents. LLMs add document interpretation, memory, planning, tool use, communication, and natural-language explanations. Synthetic-sample and laboratory-replication studies show that these agents can reproduce selected patterns, while also exposing sensitivity to prompts, model providers, and population construction [1, 4, 27]. Financial systems use these capabilities in trading, portfolio, and interacting-investor settings [20, 64]; their distinctive contribution is flexible information processing rather than automatically valid economic behavior.

2.3 Two-dimensional taxonomy

Table 2 separates two axes that system-family labels often conflate. Economic scope identifies which state and institutions are endogenous, whereas learning location identifies where a fitted or pretrained model executes; online updating during every rollout

Table 2: Core taxonomy: economic level $\times$ learning location during rollout or adaptation. Asterisks denote antecedents or supporting components outside the strict EWM boundary; daggers denote language-mediated agents. Empty cells indicate no representative in the focal corpus by the search cutoff.

Economic level	Rule-based substrate	Learned actors only	Learned environment only	Coupled learned sides
Market microstructure / assets	ABIDES [*] ; flash-crash ABM [*] ; FABS [*]	ABIDES-Gym; StockAgent [†] ; StockSim [‡]	World Agent; NS-ABM; MarS	MarS + RL; TwinMarket [‡]
Institutions / networks	Payment and contagion ABMs [*]	—	—	—
Macroeconomy	AB-SFC [*] ; DSK [*]	R-MABM; ABIDES-Economist	—	AI Economist (actor–planner)
Market–society / policy	Conventional social ABMs [*]	EconAgent [†] ; AgentSociety [‡] ; GPLab [‡]	—	PolicySim ^{‡*}

is not required. Coupling may join learned actors to a learned environment, or represent actor–institution co-adaptation, as in the AI Economist’s jointly trained agents and tax planner. The same platform can occupy several cells under different configurations.

This matrix determines the remainder of the review: Section 3 moves down the economic-level axis and compares learning locations within each level.

3 EWM Systems from Markets to Societies

Table 3 provides a cross-domain index, while the subsections below move from market microstructure through institutions and macroeconomies to market–society systems. Repository links report inspectable public artifacts located by the search cutoff; labels flag archived, partial, or simplified releases where documented. Public availability does not imply a complete licensed release, and “—” only means that no system-specific repository was identified.

3.1 Market-microstructure and asset-market EWMs

Rule-based substrates. Limit-order-book ABMs provide the most explicit institutional base. ABIDES implements a configurable exchange, message latency, and heterogeneous traders; Get Real illustrates how such worlds can be compared with observed market regularities [12, 73]. Flash-crash ABMs vary strategies and market rules under extreme conditions [30]. These systems offer controllability, but stylized background traders remain weak behavioral proxies.

Learned actors only. ABIDES-Gym and multi-agent RL studies train trading policies against a rule-governed venue [2, 40]. StockAgent and StockSim’s real-time mode place language agents in shared trading environments with executable order or market state [58, 80]. Adjacent systems such as FinCon, Agent Trading Arena, and QFinZero instead advance memory, numerical reasoning, tool use, and decision evaluation without necessarily endogenizing market response [51, 52, 78]. Machine Spirits and accidental pump-and-dump experiments show that provider choice and repeated interaction can materially change simulated market dynamics [11, 65].

Learned environment only. World Agent estimates aggregate market response while preserving an order-submission interface [17]. NS-ABM combines neural event timing and conditional order generation inside ABIDES, and MarS uses an order-level foundation model conditioned on history and incoming actions [44, 66, 67]. These models reduce hand specification of background flow, but their response under unseen assets, regimes, and interventions remains the critical uncertainty. Inverse RL, behavior cloning, and

model-based RL can improve individual components without constituting complete interactive worlds [35, 49, 68].

Coupled learned sides remain sparse: one MarS configuration adds a trained execution agent, while TwinMarket combines LLM traders with a data-trained information-recommendation component [77].

3.2 Financial-institution and network EWMs

The institution layer is rich in rule-based substrates but thin in core EWMs. Payment-system models encode settlement queues and liquidity choices, while systemic-risk ABMs represent interbank exposures, defaults, and overlapping-portfolio fire sales [13, 29]. FABS targets extensible, accelerator-supported financial-system digital twins, although its published case is market-scale rather than an end-to-end bank network [43]. Central-bank reviews show related applications in payments, housing, monetary transmission, and stress testing [9]. These models supply contracts, ledgers, and networks that learned institution-level systems must preserve.

As adjacent learned decision components, AlphaManager learns robust corporate financial controls, while ORLM generates optimization formulations [15, 34]. Neither demonstrates a learned multi-institution world. We identified no environment-only or coupled institution-level EWM that combines a learned financial network, interacting learned banks, and end-to-end accounts. This empty region is a substantive gap rather than evidence that scaling a market simulator alone yields a banking EWM.

3.3 Macroeconomic EWMs

Rule-based substrates provide the strongest accounting discipline. Agent-based stock–flow-consistent models record transactions among households, firms, banks, government, and the central bank; the DSK family extends this structure to production, finance, climate, and policy [14, 62]. Data-driven macro ABMs increasingly use microdata, calibration, and held-out forecasting [60].

Learned actors introduce endogenous adaptation. R-MABM replaces selected firms with independently trained RL policies; ABIDES-Economist exposes households, firms, government, and a central bank through a learning interface; and TaxAgent and EconGym support fiscal or multi-task experiments [10, 21, 54, 74]. RL also finds regularized equilibria in heterogeneous-agent economies [76]. **Coupled learning** is clearest in the AI Economist, a stylized mechanism-design economy that jointly adapts actors and a tax planner [82]. ABIDES-Economist is broader institutionally; neither label implies a complete empirical macroeconomy. We identified no environment-only macroeconomy in the focal corpus. These

Table 3: Representative EWM systems, institutional substrates, and adjacent systems. Placements reflect the dominant reported function, not an exclusive class or quality ranking.

Economic level	Learning location	Representative systems	Endogenous state or institution	Artifact
Market	None (substrate)	ABIDES; flash-crash ABMs [12, 30]	Exchange messages, order book, latency, inventory	ABIDES (arch.)
Market infrastructure	None (substrate)	FABS [43]	Scalable exchange / digital-twin execution	Code
Market	Actors	ABIDES-Gym; StockAgent; StockSim [2, 58, 80]	Learned or language traders in coded venues	A-Gym (arch.); S-Agent; S-Sim
Market	Environment / coupled	World Agent; NS-ABM; MarS; TwinMarket [17, 44, 67, 77]	Conditional order flow and market response	Partial; TwinMarket
Institution / network	None (substrate)	Payment and systemic-risk ABMs [13, 29]	Settlement, liquidity, exposures, contagion	–
Macroeconomy	None (substrate)	AB-SFC; DSK [14, 62]	Sectoral stocks, flows, production, finance, policy	–
Macroeconomy	Actors / coupled	R-MABM; ABIDES-Economist; AI Economist [10, 21, 82]	Firms, households, government, tax planner	Archived code
Macro / society bridge	LLM actors	EconAgent [45]	Work, consumption, income, macro feedback	Code
Market–society	LLM actors / coupled	AgentSociety; GPLab; PolicySim [37, 59, 81]	Social networks, modular subsystems, policy response	GPLab; PolicySim (simpl.)

are within-world comparisons: frozen, re-solved, adaptive, and re-estimated counterfactuals are distinct experiments [19, 63].

3.4 Market–society and policy EWMs

Conventional social ABMs provide rule-based substrates at this level, but we found no environment-only configuration. Learned language agents broaden state beyond transactions to beliefs, communication, mobility, and policy interpretation. EconAgent connects household work and consumption choices to aggregate feedback; AgentSociety supports large populations and urban-social interaction; and GPLab couples LLM decisions to modular social subsystems [45, 59, 81]. **Coupled learning** appears in PolicySim, which adapts platform intervention within social-media recommendation and opinion dynamics [37]. It supplies transferable evidence on policy–population response, but it is adjacent to financial or macroeconomic policy rather than evidence of either. These systems expand representable social state, while their economic behavior still requires comparison with external human evidence.

3.5 Cross-domain system map

Table 3 shows learned decision makers at the market, macroeconomic, and market–society levels, whereas learned institution networks and macroeconomic transition environments remain unrepresented. The map therefore exposes both a concentration of work at the market layer and the empty cells that motivate cross-level research. Architectural capability does not establish validity, however, so we next turn from where learning occurs to what evidence supports each class of economic claim.

4 Evaluation and Validation Practices Across EWMs

Evaluation requires both claim-specific criteria and empirical anchors. Table 4 catalogs representative assets used in the reviewed literature while keeping observed data, official accounts, metric suites, and agent benchmarks distinct.

4.1 Five validation dimensions

ABM validation distinguishes correct implementation from empirical adequacy for a stated purpose [18, 32, 71, 75]. Applied to EWMs, five dimensions recur. *Institutional and accounting integrity* verifies

exchange rules, contracts, settlement, budgets, and economic identities relevant to the claim. *Behavioral realism* compares conditional choices with human or institutional evidence, not merely plausible narratives. *Distributional fit* tests stylized facts and held-out outcomes. *Intervention validity* asks whether responses remain credible after strategies, rules, or policies change; a successful within-world experiment is not yet evidence that the effect transports to reality. *Reproducibility and provenance* require code, data, seeds, prompts, model versions, tool permissions, and event semantics. These dimensions summarize reported practice, not a new score or universal pass–fail test.

4.2 Metrics by economic domain

Microstructure metrics include return tails, volatility clustering, spreads, depth, order-type frequencies, inter-arrival times, cancellations, intraday profiles, market impact, and cross-asset dependence [55, 73]. Distributional distance alone is insufficient when a generator must respond to a new order or rule. **Macro and network metrics** combine stock–flow reconciliation and sectoral budget identities with held-out output, employment, inflation, credit, default cascades, exposure concentration, and welfare [13, 14, 60]. A balanced macro model may still have unrealistic actors, just as a realistic exchange need not have economy-wide accounting.

LLM-behavior metrics test rationality, numerical correctness, consistency under paraphrase, conditional response to incentives, multi-turn stability, and agreement with human decisions [50, 61]. Macro-forecast studies expose gaps that static demonstrations can miss; Agentic Economic Modeling uses small human samples to improve extrapolation; and microbehavior studies compare LLM choices with surveys before examining aggregate consequences [38, 47, 79]. Evaluation should also vary prompts, providers, population construction, and time periods to detect leakage and artificial homogeneity. Reviews find that external grounding and robustness remain less common than face validity or aggregate resemblance, while finance-specific studies call for crisis-period tests and stronger reproducibility [42, 69]. Cross-market and post-intervention evidence appears substantially thinner than within-world comparison in the systems reviewed here; forecasting reviews report a related shortage of regime and strategy-level testing [72].

Table 4: Representative data and evaluation assets. “Does not establish” identifies the principal inference that the asset alone cannot support.

Asset	Type / scope	Supports	Does not establish	Access
LOBSTER [36]	Reconstructed exchange-event / LOB data	Order-level replay, book reconstruction, empirical summaries	Counterfactual response to an unseen actor or rule	Licensed data
FI-2010 [56]	Public LOB dataset	Day-anchored mid-price prediction comparisons	Interactive market realism or cross-venue transfer	Data
Get Real [73]	Market metric suite	Stylized facts, intraday patterns, impact, dependence	Behavioral identification or response outside tested cases	Paper
LOB-Bench [55]	Evaluation suite	Distributional distances, book statistics, impact response	Institutional or behavioral validity beyond supplied tests	Code
ESA 2010 / Fed Z.1	Official accounts	Sector flows, financial transactions, asset and liability stocks	Microbehavior or counterfactual policy effects	ESA; Z.1
FRED-MD [53]	Public macro panel	Reproducible time-series and forecasting comparisons	Microbehavior, sectoral ledgers, or causal policy transport	Data
STEER [61]	LLM-choice benchmark	Fine-grained economic rationality and model comparison	Population interaction or institutional dynamics	Benchmark
Capraro et al. [16]	Human-choice data	Cross-country dictator-game distributions for LLM comparison	Repeated interaction or market behavior	Data/code
PolicySimEval [39]	Policy-simulation task benchmark	End-to-end diagnostic comparison in specified cases	Real-world policy effects outside those cases	Chapter

4.3 Data and evaluation assets

We identified no unified benchmark in the focal corpus that spans transactions, institutions, macro accounts, and language behavior. The assets in Table 4 support different claims and should not be presented as interchangeable end-to-end benchmarks.

5 Demonstrated and Prospective Applications

5.1 Demonstrated experimental uses

The strongest evidence concerns within-simulator research workflows rather than production deployment. Demonstrated cases include order-execution and market-impact experiments in ABIDES-Gym and MarS, together with flash-crash and market-design scenarios in mechanically explicit venues [2, 30, 44]. Institutional ABMs examine payment and contagion mechanisms [13, 29]. Stylized economies compare tax, production, and household-labor settings [21, 45, 82], while PolicySim studies platform-recommendation and opinion interventions [37]. These studies establish experimental utility, not corresponding real-world profit, welfare, or policy effects.

5.2 High-stakes uses requiring further validation

Bank-wide credit, funding, collateral, payment, and liquidity-risk worlds remain prospective because these elements are not yet jointly represented with validated learned responses in the systems reviewed here. Cross-region macro transmission poses the same problem at a larger scale. For regulators, systemic-risk EWMS could compare contagion, correlated liquidation, or rule changes, but current evidence supports transparent scenario laboratories rather than policy oracles [9, 13]. The Financial Stability Board identifies concentration, correlated decisions, cyber dependence, model risk, and data quality as possible AI amplification channels [28]; ECB architecture experiments and BIS Project Logos provide early controlled investigations rather than deployment validation [3, 8].

5.3 Stakeholder requirements

Quantitative firms need realistic costs, latency, market impact, leakage-resistant time splits, and regime tests. Central banks need accounting identities, heterogeneous balance sheets, monetary and fiscal transmission, and uncertainty around counterfactuals. Regulators need traceable exposures, manipulation and concentration tests, adverse scenarios, and auditable provenance. Academic users need public configurations, modular baselines, negative results, and comparable metrics. LLM agents add document processing and coordination, but numerical reliability and live-market evidence remain unresolved [23, 64]. AlphaManager and ORLM illustrate adjacent corporate-control and optimization components [15, 34]. Public repositories and within-world tests are visible for several systems, whereas the reviewed literature provides comparatively little independent replication, live-data integration, or documented operational deployment.

6 Common Challenges and Research Frontiers

The gap between these application requirements and present evidence motivates the limitations and research tracks below.

6.1 Six recurring limitations

Across the reviewed systems, six limitations recur. **Behavioral identification bias** arises when plausible personas or matched aggregates conceal incorrect conditional decisions. **Weak cross-market generalization** limits transfer across assets, venues, regions, and regimes. **Missing accounting integration** prevents local actions from propagating through contracts and balance sheets. **Incomplete reproducibility** follows from unavailable data, proprietary models, changing APIs, and underreported prompts or seeds. **LLM hallucination and numerical error** can create unsupported beliefs, inconsistent actions, or invalid quantities inside otherwise executable worlds. **Deployment cost** grows with event frequency, population size, long context, repeated seeds, and human oversight. Larger agent counts do not resolve these problems when actors share a model provider, prompt template, or training corpus [28, 42, 69].

Table 5: Expanded technical index: economic level $\times$ learning location. A: rule substrate; S: offline/component support; J: outside-boundary work; †: language-mediated. Unmarked entries meet the EWM definition; data/metrics are in Table 4.

Economic level	Learning location	Systems and methods
Cross-level	Offline support	Parameter calibration; data assimilation; Bayesian optimization; gradient-assisted calibration; RL-combined search; synthetic-sample and laboratory replication; Homo Silicus; Agentic Economic Modeling; LLM microbehavior alignment ^S
Market microstructure / assets	None	ABIDES ^A ; flash-crash ABMs ^A ; FABS ^A
Market microstructure / assets	Learned actors	ABIDES-Gym; LOB multi-agent RL; StockAgent [†] ; StockSim real-time mode [†] ; Machine Spirits [†] ; accidental pump-and-dump experiment [†] ; Project Logos ^{†J}
Market microstructure / assets	Actor-side support / adjacent	In-Trajectory IRL ^S ; attention-based behavioral cloning ^S ; model-based RL ^S ; FinCon ^{†J} ; Agent Trading Arena ^{†J} ; QFinZero ^{†J}
Market microstructure / assets	Learned environment	World Agent; state-dependent neural Hawkes process ^S ; NS-ABM; MarS
Market microstructure / assets	Coupled learning	MarS + RL execution agent; TwinMarket [†]
Institutions / networks	None	Payment-system ABM ^A ; systemic-risk and contagion ABMs ^A
Institutions / corporate finance	Learned decision support	AlphaManager [†] ; ORLM ^{†J}
Macroeconomy	None / offline support	AB-SFC ^A ; DSK ^A ; data-driven macro ABMs with microdata calibration and held-out forecasting ^S
Macroeconomy	Learned actors	R-MABM; ABIDES-Economist; TaxAgent [†] ; EconGym; regularized-equilibrium RL ^S ; synthetic-persona macro forecasting ^{†J}
Macroeconomy	Coupled learning	AI Economist actor-planner
Market-society / policy	None / learned actors	Conventional social ABMs ^A ; EconAgent [†] ; AgentSociety [†] ; GPLab [†]
Market-society / policy	Coupled learning	PolicySim ^{†J}

6.2 Five research tracks

Five research tracks respond directly to those limitations. **Multi-scale coupled simulation** should connect order flow, institution balance sheets, and macro demand so that shocks propagate across levels. **Accounting-native hybrid EWMs** should let learned actors propose decisions while deterministic layers enforce contracts, settlement, units, ownership, and sectoral identities. **Unified intervention benchmarks** should separate interpolation from new regimes and compare frozen, adaptive, re-solved, and re-estimated worlds with placebo and adverse interventions. **Lightweight open frameworks** should provide governable APIs, versioned configurations, replayable traces, cost reporting, and provider-independent baselines. **Human-AI collaborative simulation** should use domain experts and small, high-quality human samples to design scenarios, correct behavior, inspect causal narratives, and set escalation boundaries [47, 79]. Together these tracks shift progress from larger synthetic populations toward more identifiable, accountable, and reusable experiments.

7 Limitations of This Survey

This structured narrative survey is selective rather than exhaustive, and English-language discovery may omit adjacent or non-English work. The source base mixes archival publications with preprints and institutional projects whose status may change. Taxonomy placement reflects a system’s dominant reported component rather than an exclusive class. We provide neither a meta-analysis nor independent replications; recurring patterns apply to the reviewed corpus and are not prevalence estimates or deployment audits.

8 Conclusion

Economic world models do not replace agent-based economics; they combine learned behavior or environments with executable economic systems. Market simulators provide transaction mechanics, stock-flow models accounting discipline, learned generators adaptive responses, and LLM agents richer information processing. No lineage reviewed here combines all these strengths.

Progress is therefore integrative rather than scale-driven. Larger populations and fluent narratives matter only with explicit institutions, empirically grounded behavior, regime-aware evaluation, and

reproducibility. Under those conditions, EWMs can become useful laboratories for market, risk, and policy research while keeping the boundary between simulated evidence and real-world decisions visible.

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